CARELOOP

(AI-Powered Treatment Outcome Intelligence)

TEAM
Escape 🚀

PROBLEM STATEMENT
PS5 — Open Innovation

HACKATHON
HackCrux, LNMIIT Jaipur

YEAR
March 2026

Rishi · Ritesh Kumar · Anshuman Singh · Shivam Pareek

PROBLEM STATEMENT

Healthcare Has No Feedback Loop

Industr y	Complete Journey Tracked
🍔 Food Delivery	Order placed → Preparing → Out for delivery → Delivered ✓
📦 E-commerce	Ordered → Shipped → In transit → Delivered → Review ✓
✈️ Travel	Booked → Check-in → Boarding → Landed ✓
🏥 **Healthcare**	**Doctor prescribes → ??? → Patient shows up weeks later**

```
Doctor Visit #1                          Doctor Visit #2
    | ← ← ← ← ← ← ← COMPLETE SILENCE → → → → → → → |
Did patient buy medicines?                ❓ UNKNOWN | |
Did they take them daily?                 ❓ UNKNOWN | |
Did side effects appear?                  ❓ UNKNOWN | |
Did the condition improve?                ❓ UNKNOWN | |
▼ ▼ PRESCRIPTION WRITTEN "How are you feeling now?"
```

> *The doctor is forced to start every visit* ***completely blind*** *— relying entirely on patient memory, which is unreliable.*

WHY IT MATTERS

The Numbers Are Alarming

50%
Prescriptions not properly followed

$500B
Annualglobal cost ofnon-adherence

30%
Hospital readmissionslinked to non-adherence

5–8 min
Avg. doctor time per outpatient visit

Doctors
Start every visit with zero context. Ask "Did you take your medicines?" and hope for an honest answer.

Patients
Forget schedules, can't recall side effects, lose paper prescriptions, miss follow-ups.

Healthcare System
Wastes billions on preventable readmissions, duplicate tests, and failed treatment cycles.

India alone: **101 Million+ diabetics** *and* **220 Million+ hypertensive patients** *— all requiring continuous monitoring between visits. Nobody is tracking what happens between those visits.*

MARKET ANALYSIS

Everyone Is Fighting Over the Front Door

We researched **40+ funded healthcare startups** across Y Combinator (2020–2026) and India's ecosystem. The finding was unanimous:

Platform	Records	Adherence Tracking	Treatment Plan AI	Outcome Measurement	Between-Visit Intel
Eka Care	✓	✗	✗	✗	✗
Practo	✓	✗	✗	✗	✗
Tata 1mg	✗	✗	✗	✗	✗
Prana (YC)	✓	Partial	✗	✗	✗
CareLoop ✦	✓	✓	✓	✓	✓

> **Every startup is fighting over the front door of healthcare. Nobody is building for what happens after the patient walks out.**

PROPOSED SOLUTION

Introducing CareLoop — Outcome Engine™

> *CareLoop transforms every prescription from a* **dead piece of paper** *into a* **living, intelligent treatment journey** *— and closes the feedback loop that healthcare has never had.*

✕ CareLoop is NOT

- → A booking or appointment app
- → A static health record vault
- → A chatbot or call bot
- → A hospital management system

✅ CareLoop IS

- → A **Treatment Outcome Intelligence Engine**
- → A system that measures if treatments **actually work**
- → A bridge across the **silence between visits**
- → An **AI copilot** for patients and doctors

SYSTEM OVERVIEW

The 4-Layer CareLoop Outcome Engine™

Layer 1 — Smart Rx Engine

Parses any prescription (handwritten or digital) via OCR + LLM. Generates **expected outcomes**, side-effect watch list, follow-up lab schedule, and success criteria.

Layer 2 — Between-Visit Intelligence

1-tap daily adherence check-ins · Guided side-effect logging · Symptom tracking · Lab report upload & auto-parsing — all between visits.

Layer 3 — Treatment Outcome Report

Before the next visit, auto-generates a complete **Treatment Outcome Report**: adherence scores, lab value trends, effectiveness rating, and suggested consultation focus areas.

Layer 4 — Emergency Health QR

Scannable QR card showing blood group, allergies, active medications with adherence scores, and key lab values — accessible by any doctor in seconds during an emergency.

PRESCRIPTION UPLOADED → Smart Plan Generated → Between-Visit Tracking → **Outcome Report Ready Before Next Visit**

LAYER 1 — SMART PRESCRIPTION ENGINE

From Raw Prescription → Intelligent Treatment Plan

Doctor writes: *"TabMetformin500mgBD×30days.FBSandHbA1cafter3months."*

Component Generated	Output
💊 Medications	Metformin 500mg — Twice daily — 30 days
⚠️ Side-Effect Watch	GI discomfort, nausea (common, expected in first 2 weeks)
🔬 Follow-up Labs	FBS + HbA1c — Due: June 15, 2026
🎯 Success Criteria	HbA1c < 7.0 · FBS < 130 mg/dL
📅 Next Visit	~3 months after lab results are uploaded

> *For the first time, a prescription has* **defined expected outcomes** *and* **measurable success criteria** *— before the patient even leaves the clinic.*

LAYER 3 — THE INNOVATION

Treatment Outcome Report — What the Doctor Sees

```
📊 TREATMENT OUTCOME REPORT

   Patient: Rahul Kumar, 45M
   Period: March 15 → June 10, 2026
   Generated by CareLoop Outcome Engine™
   ──────────────────────────────────────

   OVERALL EFFECTIVENESS: 62%
   ▓▓▓
```

TECHNICAL ARCHITECTURE

Technologies Used

Component	Technology	Purpose
Prescription OCR	Google Cloud Vision + LLM	Handwritten / printed Rx → structured treatment plan
Treatment Plan AI	Gemini / GPT-4	Generate outcomes, success criteria, side-effect watch
Backend	Node.js / Python FastAPI	API layer, business logic, data orchestration
Database	PostgreSQL + Redis	Patient data, treatment timelines, caching
Patient Mobile App	React Native / Flutter	1-tap adherence tracking, side-effect logging
Doctor Dashboard	React.js	Outcome reports, case review, alert management
Lab Report Parser	OCR + NLP Pipeline	PDF lab reports → structured values with trend analysis
Interoperability	FHIR / ABDM Standards	India's national digital health infrastructure compatibility

WHY NOW

Four Forces Have Converged in 2026

Force	What Changed	Impact on CareLoop
🏗️ ABDM Rails Maturing	India's ABHA IDs and FHIR standards are live and scaling.	A unified patient identity exists for the first time — we build on top of it.
🧠 LLMs for Longitudinal Reasoning	GPT-4, Gemini can now analyze months of patient data and reason across it.	A Treatment Outcome Report was impossible before 2023. Now it's one API call.
📱 800M+ Smartphones in India	Even tier-2/3 cities have smartphone penetration.	The patient input layer — 1-tap adherence tracking — is finally viable at scale.
📈 Chronic Disease Epidemic	101M+ diabetics, 220M+ hypertensive — growing 10%+ annually.	Chronic disease requires continuous monitoring. That's exactly what we do.

> *"CareLoop isn't early or late — it's* **exactly on time**.*"*

COMPETITIVE ADVANTAGE

Our Moat Grows With Every Patient

Feature	Others	CareLoop
Record Storage	✓	✓
Prescription OCR	✓	✓
Adherence Tracking	✗	✓
Treatment Plan Generation	✗	✓
Outcome Measurement	✗	✓
Between-Visit Intelligence	✗	✓
Emergency Health QR	✗	✓

📊 Data Moat

Every treatment cycle generates unique outcome data — no competitor can catch up without years of patient data.

🧠 Behavioral Moat

Adherence patterns become training data for smarter nudge predictions and outcome models.

🔒 Switching Cost

2+ years of curated treatment history is irreplaceable for patients, doctors, and health systems.

ANTICIPATED IMPACT & DEVELOPMENT PLAN

Measurable Outcomes & Agile Execution

40–50%
Non-adherence events flagged earlier

5–10 min
Doctor time saved per visit

30%
Preventable readmissions addressed

30 sec
Emergency health context via QR scan

Phase	Timeline	Key Deliverables
Planning	Day 1 AM	Architecture finalized · UI mockups · Synthetic dataset · Repo + cloud setup
Development	Day 1–2	Prescription OCR · LLM Outcome Engine · Patient adherence tracker · Doctor dashboard
Testing	Day 2–3	End-to-end flow testing · Edge case handling · AI output quality validation
Polish	Day 3	UI refinement · Demo script prep · Rehearsal (3×) · Backup demo video

OUR VISION

CareLoop

"In every industry,we

trackoutcomes.
We track if your food arrived, if your package was delivered,

if your flight is on time.

But in healthcare — the most critical

industry of all —

nobody tracks if your treatment actually worked.
CareLoop changes that."

RISHI · RITESH KUMAR · ANSHUMAN SINGH · SHIVAM PAREEK

HackCrux · LNMIIT, Jaipur · March 2026 · PS5 — Open Innovation